

Good comparison to make — iFAST is the most directly relevant published work, and it's a genuinely different sensing philosophy from what I designed for you.

****iFAST 2020 (Spencer et al., Nature Communications)**** is single-cell microfluidic impedance cytometry, not bulk-well impedance. Bacteria flow one at a time through a microfluidic channel where high-frequency impedance measurements capture the electrical and morphological properties of many thousands of individual cells at high throughput [PubMed](<https://pubmed.ncbi.nlm.nih.gov/40623549/>) . The parameters extracted per cell are electrical radius as a proxy for cell size and opacity as a proxy for membrane integrity, both of which shift in susceptible cells but not resistant or untreated ones [360Dx](<https://www.360dx.com/diagnostics/ifast-aims-deliver-three-hour-antimicrobial-susceptibility-test-results-impedance-based>) . Timing is remarkable: after an overnight culture, cells are revived for 30 minutes, exposed to antibiotic for another 30 minutes, then measured in just 3 minutes [PubMed](<https://pubmed.ncbi.nlm.nih.gov/40623549/>) — because it's detecting near-instant membrane/morphology damage in individual cells rather than waiting for population-level metabolic or growth changes.

****iFAST 2025 (Journal of Infection, UTI Enterobacterales study)**** is the same underlying platform pushed into clinical validation. Fifty-eight *E. coli* and *K. pneumoniae* strains were exposed to breakpoint antibiotic concentrations, and after two hours the drop in single-cell count relative to an unexposed control was used to call susceptibility [ScienceDirect](<https://www.sciencedirect.com/science/article/pii/S0163445325001434>) . For MIC-level resolution, strains near breakpoint were run against doubling-dilution antibiotic series to derive an electrical MIC, giving 100% correlation with broth microdilution on lab strains within a five-hour test, and clinical concordance in at least 74 of 80 real patient samples [nih](<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9848847/>) .

****Where your design differs — three real gaps, not just cosmetic ones:****

- ****Signal type.**** iFAST reads individual-cell electrical morphology at high frequency (MHz range, size/shape/membrane damage). Your prototype reads aggregate medium conductivity at low frequency (100 Hz–10 kHz) — a population-level metabolic proxy, closer

to classical impedance microbiology (Bactometer/RABIT) than to iFAST. This is the fundamental reason iFAST hits 30–60 minutes and yours realistically lands at 30 min–3 h: single-cell membrane damage is visible almost immediately, while bulk conductivity has to wait for enough cells to collectively shift the medium's ion content.

- **Hardware class.** iFAST needs a microfluidic flow-cytometry-grade impedance system with precise single-cell flow control and multi-frequency lock-in detection — instrument-grade, not a \$200–700 bench build. Your AD5941 + static-well IDE design is deliberately in a completely different cost/complexity tier; it can't and isn't trying to do single-cell resolution.

- **Validation maturity.** iFAST 2020 was proof-of-concept across species (also demonstrated on MRSA/MSSA and *N. gonorrhoeae* in follow-up conference work); iFAST 2025 is a clinical concordance study on real patient samples. Your design is still at the bench-prototype/protocol stage — the whole point of the staged validation plan (ATCC 25922 vs 35218, then dose-response, then eMIC regression) is to earn the same kind of evidence iFAST already has.

Where the two converge nicely: the electrical-MIC concept — dosing across a two-fold dilution series and regressing an electrical readout against reference broth-microdilution MIC — is exactly the same logic in both iFAST 2025 and your DIR/eMIC framework. That's good validation that your threshold methodology is sound; you're just applying it to a cheaper, slower, coarser signal (bulk conductivity) instead of iFAST's expensive, fast, fine-grained one (single-cell electrical phenotype).

There's also a commercialized descendant worth knowing about: a 96-well plate version incubates enriched patient samples with freeze-dried antibiotics at multiple concentrations per well and uses multi-frequency impedance to detect bacterial volume changes and cell-wall permeability changes [Journal of Infection](<https://www.journalofinfection.com/article/S0163-4453%2825%2900143-4/fulltext>) , aiming for full AST-plus-MIC panels in about three hours — effectively an automated, clinical-grade instrument built on the same physics, which underscores that iFAST's real product isn't just the sensor, it's the whole integrated culture-to-result pipeline your low-cost version deliberately skips.